Temperature Control, Hard Problem

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This is a control theory problem. The idea is to control the temperature of an object that is constantly dissipating heat, and also receives some noise from the environment.

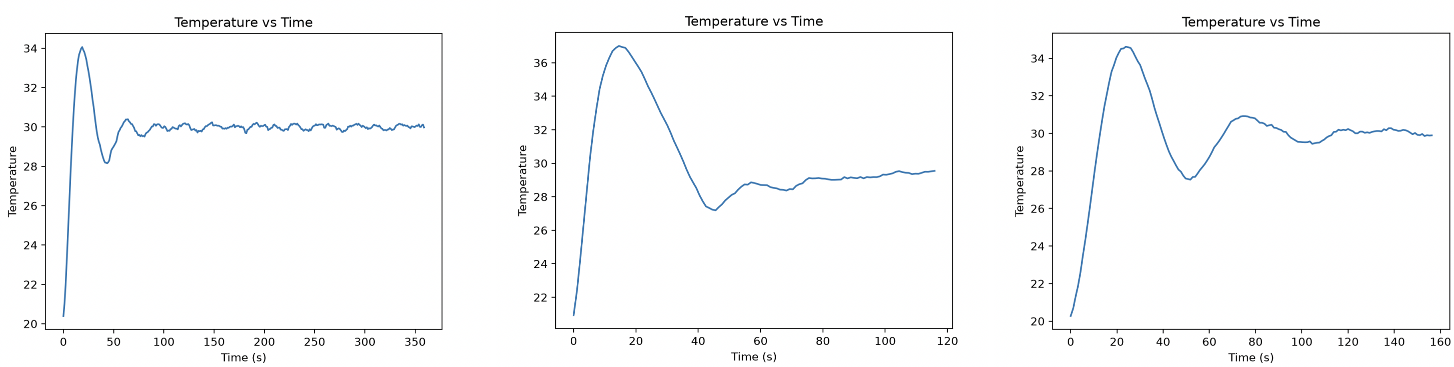
The Story
You have a houseplant named Pickles. Pickles, unfortunately, can only survive at 40 degrees Celsius (not above, not below). The house is at room temperature, which for us will be 20 Celsius. As a solution, you build Pickles a glass box, which partially seals him off from the environment. The only interaction between Pickles and outside of the box is thermal dissipation.

To heat this box, you install a heater, but you find that after running the heater for a while, Pickles is at 60C: It is like an oven! You need a better solution. You need a control algorithm.

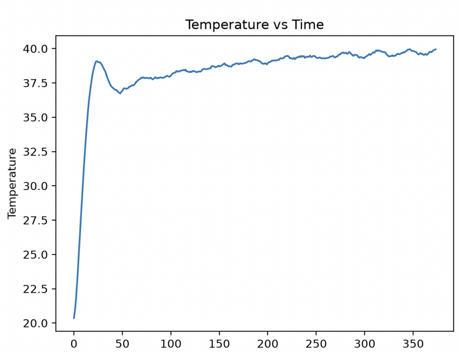
You install a temperature sensor inside the box as the input to your control algorithm. You also upgrade the heater to a heat pump, which can both heat and cool the box as needed. Your task is simple:

Design a function which takes in the temperature of the box and the setpoint (in our case 40C) and outputs a control signal to the heat pump (0 being off, positive being heat, and negative being cool). The specific scaling for this control signal is an unknown, and is accounted for as part of your tuning process for the algorithm you create.

The starter code for this problem is in the same folder that you found this PDF. Please download this code through whatever method you wish (I suggest using git clone on the repository) and implement the controller function in Controller.py.
As I note in the comment above this function, you very likely will need variables outside in order to preserve state between calls. This is totally fine, and is more or less expected.



Above: Three different **poor** control algorithms. Note the high overshoot and oscillations



Above a better (but not perfect) control algorithm. Note no overshoot and less oscillations